EMPLOYEE PRODUCTIVITY ANALYSIS USING REGRESSION MODELS

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1. INTRODUCTION

This project analyzes the factors affecting employee productivity using statistical methods and regression analysis.

The objective is to identify how experience, salary, overtime and department influence productivity and to build a predictive model.

The dataset used in this analysis is simulated for analytical purposes.

2. VARIABLES

Dependent Variable:

• Productivity Score (y)

Independent Variables:

• Experience (x1)

• Salary (x2)

• Overtime (x3)

• Department (x4 – categorical)

3. DATA PREPARATION

Initial analysis showed that the dependent variable did not follow a normal distribution.

A logarithmic transformation was applied:

$\ln(y) = \log(y)$

After transformation, the data satisfied the normality assumption ($p > 0.05$).

4. REGRESSION MODEL

A multiple linear regression model was constructed:

$\ln(y) = \beta 0 + \beta 1x1 + \beta 2x2 + \beta 3x3 + \beta 4x4$

Estimated model:

Ln(y)=3.2510+0.14xi+0.081x2-0.039x3-0.099x42-0.123x43 +-0.030

(0.056) (0.004) (0.0052) (0.0055) (0.010) (0.0094)

The model was found to be statistically significant ($p < 0.05$).

5. KEY FINDINGS

• Experience has a positive and significant effect on productivity

• Salary positively affects productivity

• Overtime has a negative impact (possible burnout effect)

• Department differences significantly influence productivity

After data cleaning, the model explains approximately 97% of the variance ($R^2 \approx 0.97$).

6.MODEL DIAGNOSTICS

Several tests were conducted to validate model assumptions:

• Multicollinearity: No issue detected (VIF < 5)

• Heteroscedasticity: Detected (Breusch-Pagan test)

• Autocorrelation: Detected (Durbin-Watson test)

These results indicate that while the model is strong, some assumptions are partially violated.

7.OUTLIER ANALYSIS

Outliers were detected using:

• Cook's Distance

• Leverage values

• Standardized residuals

After removing influential observations, model performance improved significantly.

8.MODEL IMPROVEMENT

Forward, backward and stepwise selection methods were applied.

All methods resulted in the same final model, indicating that all variables contribute significantly.

Ridge regression was also applied and confirmed the stability of the model.

9.PREDICTION

The model was used to estimate productivity scores for both existing and new observations.

Prediction intervals were calculated to evaluate uncertainty.

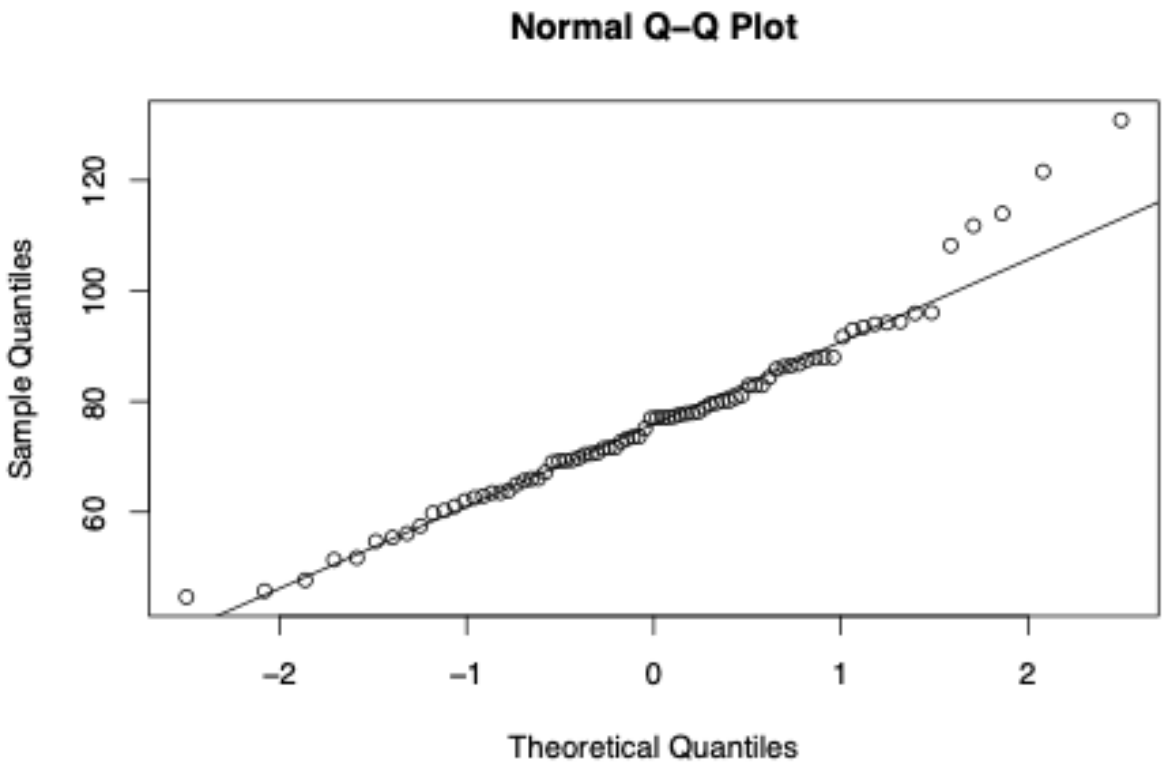
10.CONCLUSION

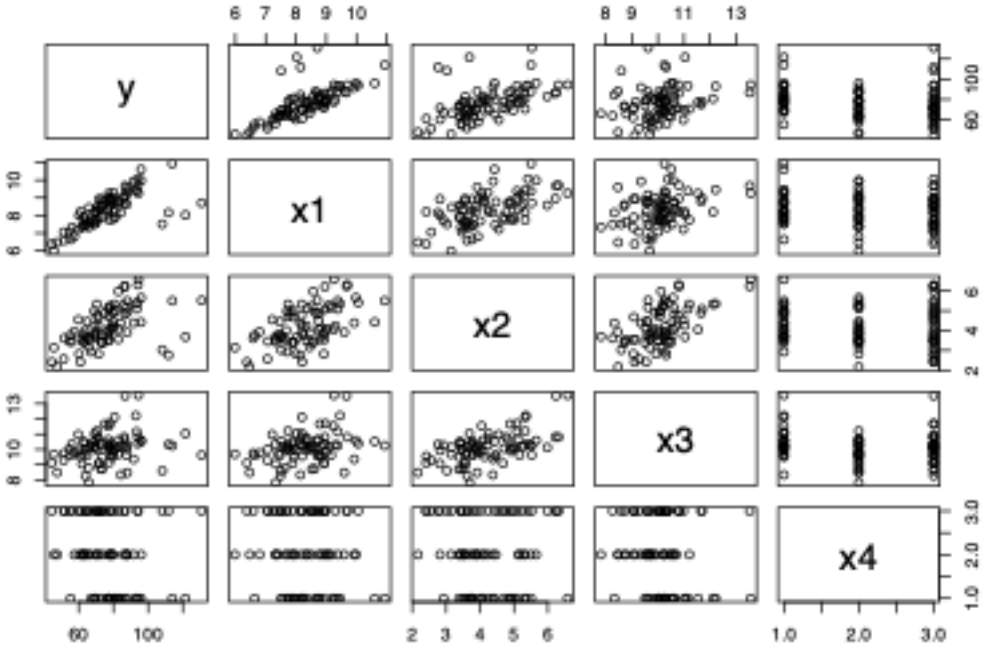
This analysis shows that employee productivity is strongly influenced by experience, salary and workload.

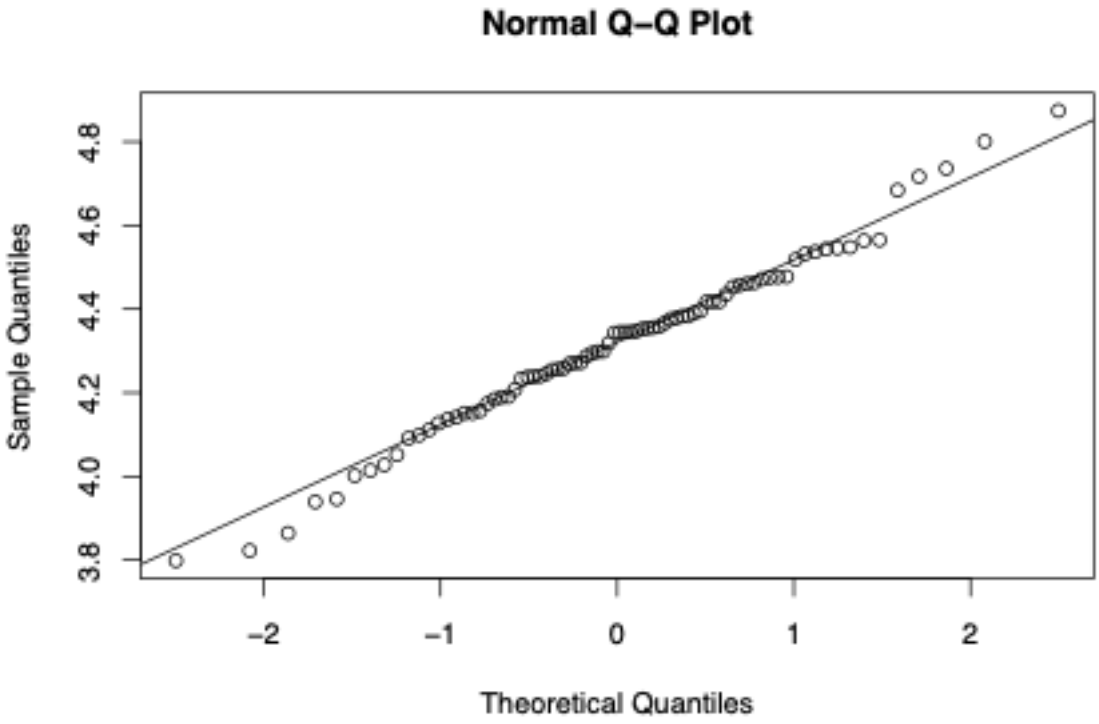
While the model performs well, further improvements could be made by including additional variables such as job satisfaction or education level.

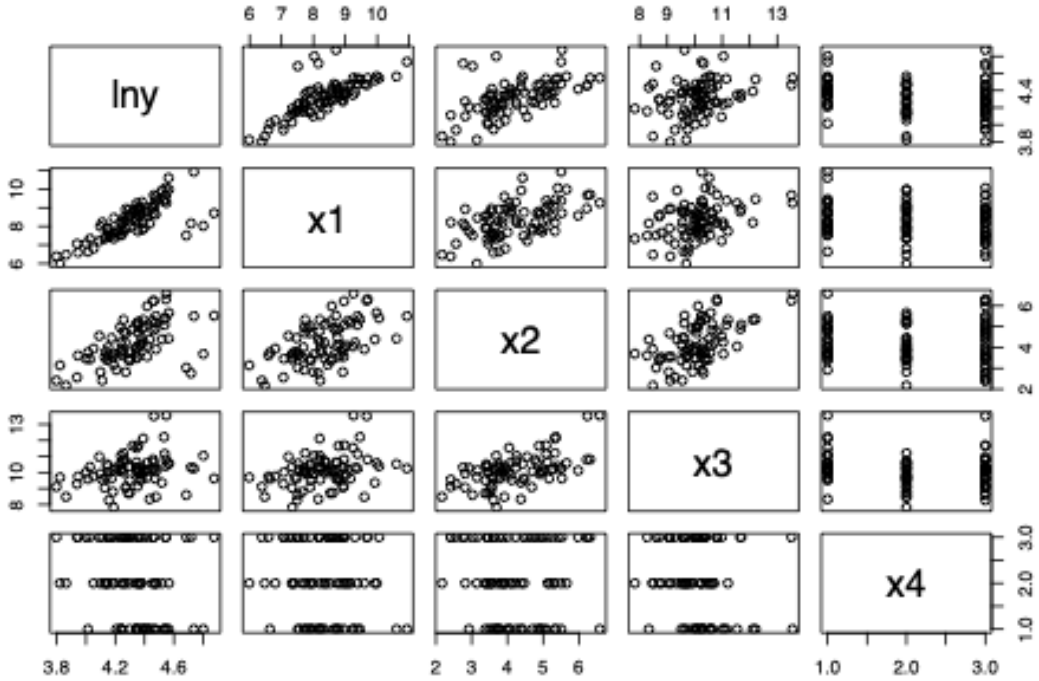
This project demonstrates the application of statistical modeling techniques in real-world decision-making scenarios

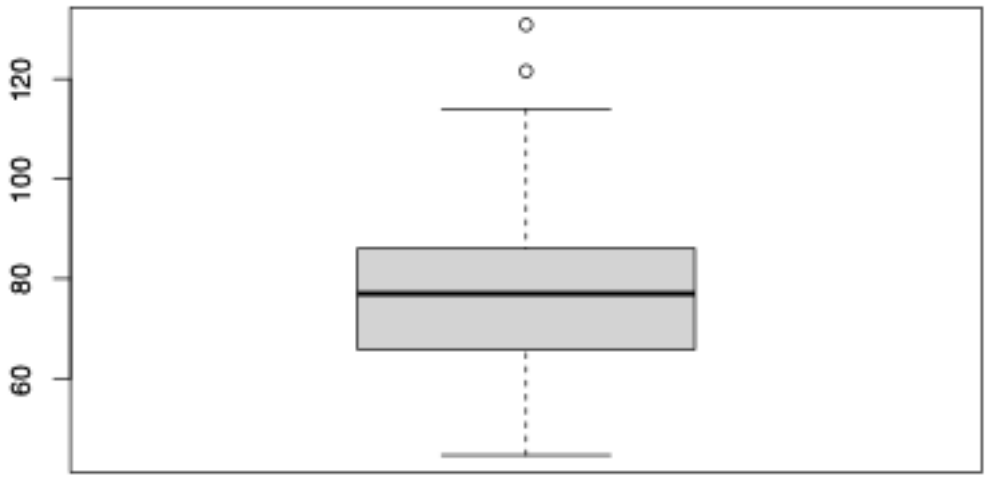
This model can be used as a decision-support tool for workforce optimization.

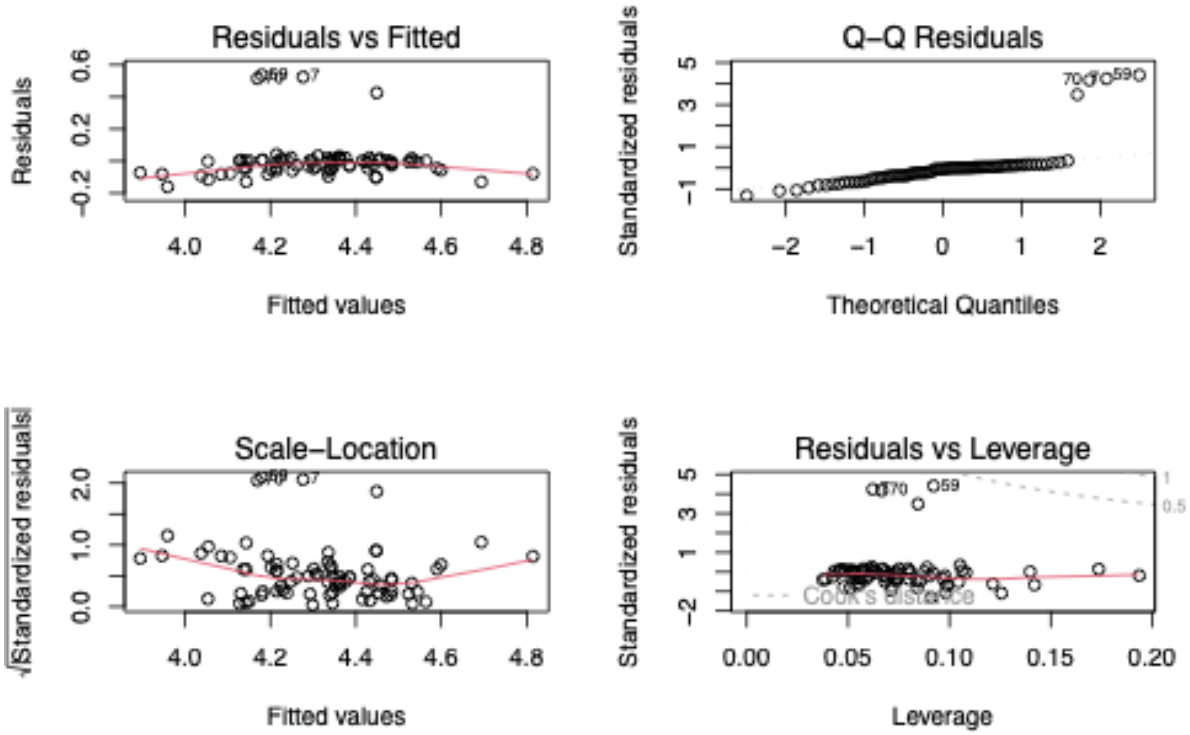

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
60
80
100
120
−2
−1
0
1
2

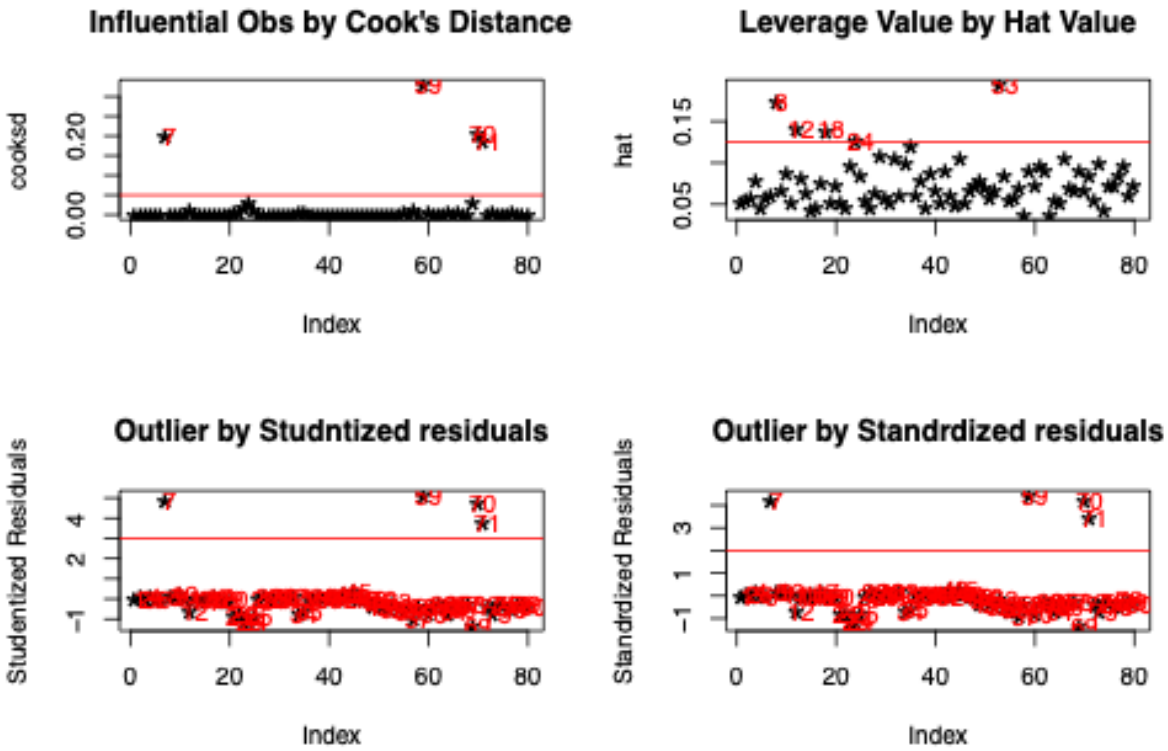

y
x1
x2
x3
x4

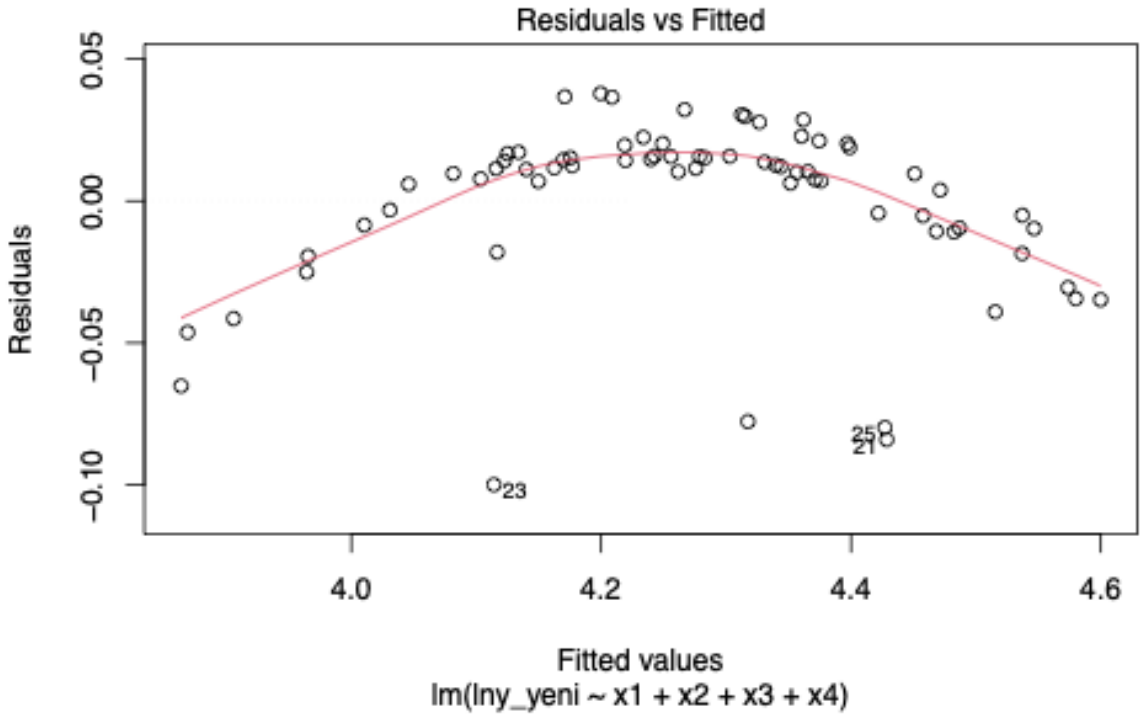

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
3.8
4.0
4.2
4.4
4.6
4.8
−2
−1
0
1
2

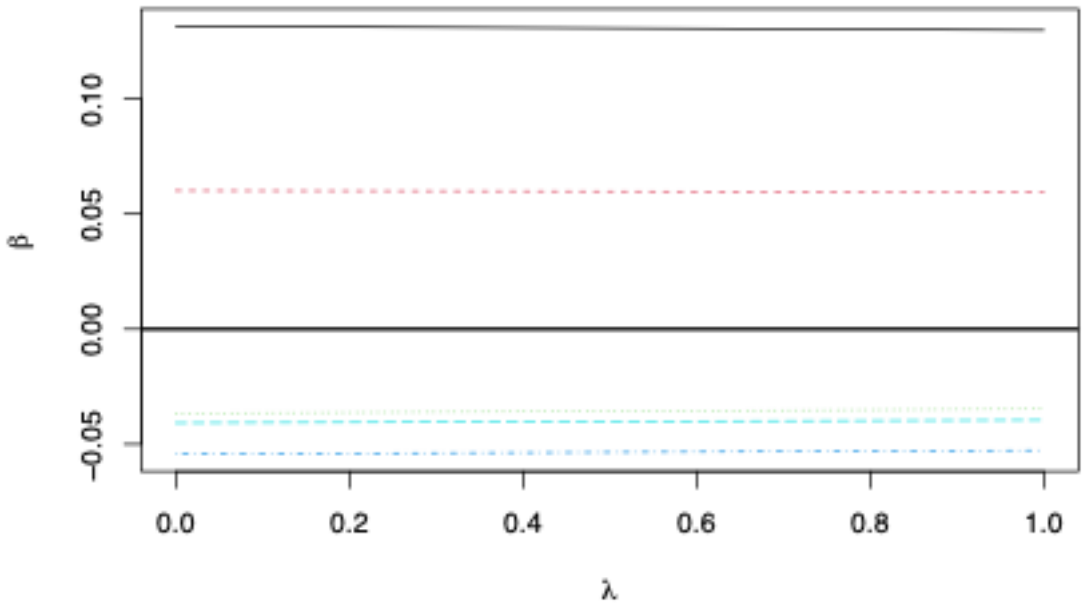

lny
x1
x2
x3
x4
6 7 8 9 10
8 9 11 13
3.8 4.4
6 8 10
2 4 6
8 10 13
1.0 2.0 3.0
3.8 4.2 4.6
2 3 4 5 6
1.0 2.0 3.0


120
100
80
60


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage


Influential Obs by Cook's Distance
cooksd
Index
Leverage Value by Hat Value
hat
Index
Outlier by Studntized residuals
Studentized Residuals
Index
Outlier by Standrdized residuals
Standrdized Residuals
Index


Residuals vs Fitted
Residuals
Fitted values
lm(lny_yeni ~ x1 + x2 + x3 + x4)


β
0.10
0.05
0.00
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
λ